Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	05 March 2026
Team ID	MCA_DA2
Project Name	Plugging into the Future: An Exploration of Electricity Consumption Patterns
Maximum Marks	4 Marks

Technical Architecture:

The deliverable includes the architecture overview and the two tables below.

Architecture Flow (3-Tier)

[User Browser] ⟷ [Flask Web App] ⟷ [Tableau Public (embed)] ⟷ [Tableau Desktop] ⟷ [MySQL 8.0] ⟷ [Consumption.csv]

Table 1 — Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Web dashboard in browser — no software installation needed	HTML5, CSS3, Bootstrap 5, JavaScript
2.	Web Framework	Handles routing and renders Tableau embeds in HTML templates	Python Flask 2.x, Jinja2 Templates
3.	Business Intelligence	All visualisations, dashboards, and storyboard	Tableau Desktop 2023 / Tableau Public
4.	Database	Stores all 16,600 rows — runs SQL aggregations	MySQL 8.0, MySQL Workbench
5.	Data Processing	Validates CSV, loads data to MySQL	Python 3.x, Pandas, mysql-connector-python
6.	Embed Layer	Tableau dashboards embedded via iframe in Flask	Tableau Public Embed / HTML iframe
7.	Version Control	Source code on GitHub — team collaboration	Git, GitHub
8.	Credentials	DB passwords stored in .env — not hardcoded	python-dotenv

Table 2 — Application Characteristics:

S.No	Characteristic	Description	Technology
1.	Open-Source	All tools free and open-source (Tableau Public for publishing)	Flask (BSD), MySQL Community (GPL), Bootstrap (MIT), Pandas (BSD)
2.	Security	DB credentials in .env file. .env excluded from GitHub. No user data collected.	python-dotenv, .gitignore, Flask DEBUG=False
3.	Scalable	New years of data: just INSERT rows. Tableau extract auto-refreshes. Flask supports new tabs.	MySQL append-only, Tableau extract refresh
4.	Availability	Tableau Public available 24/7. Flask starts with one command.	Tableau Public CDN, python app.py
5.	Performance	Tableau extract (.hyper) for fast loads. MySQL indexes on Dates + Regions.	Tableau .hyper extract, MySQL CREATE INDEX